

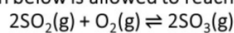


rates of reaction

Name:

Date:

The reaction below is allowed to reach equilibrium:



Explain why a high pressure yields more sulfur trioxide (SO_3).

- A. The reaction favours the side with the fewer molecules to reduce the pressure and there are less product molecules than reactant molecules.
- B. The reaction favours the side with more molecules to increase the pressure and there are more product molecules than reactant molecules.
- C. The reaction favours the side with the fewer molecules to reduce the pressure and there are more product molecules than reactant molecules.

Correct Answer: A B C D

Explanation:

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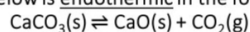
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The reaction below is endothermic in the forward direction:



What will be the effect of increasing the temperature on the reaction?

- A. Equilibrium position shifts left. The rate of reverse reaction increases to heat the reaction so more CaCO_3 is produced.
- B. Equilibrium position shifts right. The rate of forward reaction increases to cool the reaction so more CaCO_3 and CO_2 is produced.
- C. Equilibrium position shifts left. The rate of reverse reaction increases to cool the reaction so more CaCO_3 is produced.
- D. Equilibrium position shifts right. The rate of reverse reaction increases to cool the reaction so more CaCO_3 and CO_2 is produced.

Correct Answer: A B C D

Explanation:

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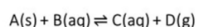
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The reaction below is set up in a closed vessel and allowed to reach dynamic equilibrium:



A valve is opened and some of D is allowed to escape. How will this affect A and B?

- A. The system will oppose the change by increasing the amount of C and D.
- B. The system will oppose the change by decreasing the amount of C and D.

Correct Answer: A B C D

Explanation:

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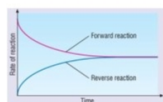
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Why does the forward rate of reaction decrease on this graph?



A. As the reaction progresses, the concentration of reactants increases. Therefore there are more collisions between reactant molecules.

B. As the reaction progresses, the concentration of reactants decreases. Therefore there are fewer collisions between reactant molecules.

Correct Answer: A B C D

Explanation:

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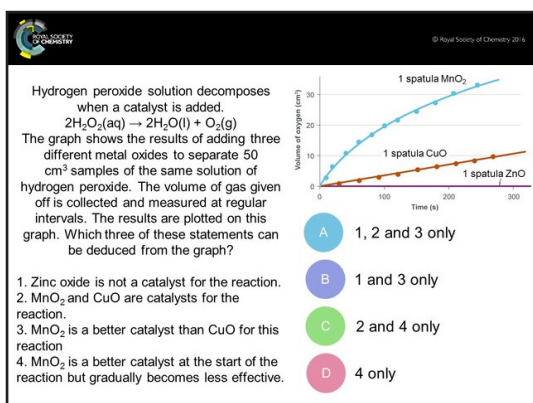
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Correct Answer: A B C D

Explanation:

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When H₂ and N₂ react they form NH₃ in a reversible reaction:
 $\text{N}_2(\text{g}) + 3\text{H}_2(\text{g}) \rightleftharpoons 2\text{NH}_3(\text{g})$
 The forward reaction is exothermic

Which combination of conditions will yield the greatest amount of NH₃?

A. High temperature and low pressure.

B. High temperature and high pressure.

C. Low temperature and high pressure.

D. Low temperature and low pressure.

Correct Answer: A B C D

Explanation:

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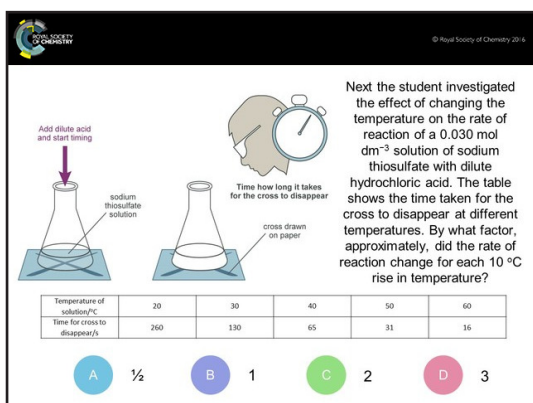
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Correct Answer: A B C D

Explanation:

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The table shows the results of three experiments to investigate the effect of changing concentration, c , on the rate of reaction of marble chips and dilute hydrochloric acid. At the start of the each reaction, the investigator measured the time, t , for a certain small volume, x , of gas to be collected. The investigator used the results to calculate a value for the rate of reaction at the start of each experiment. Which formula was used to calculate the rate?

Concentration, c , of dilute HCl(aq)/ mol dm^{-3}	Time t to collect $x \text{ cm}^3$ of carbon dioxide/s
2.0	8
1.0	16
0.5	32

A rate = $\frac{x}{t}$ B rate = $\frac{t}{x}$
 C rate = $\frac{c}{t}$ D rate = $\frac{t}{c}$

Correct Answer: A B C D

Explanation:

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Two experiments were carried out to investigate the effect of changing the acid concentration on the rate of reaction of marble chips with dilute hydrochloric acid. In the first (original) experiment the volume of gas given off was measured over time on mixing 100 cm^3 of 2.0 mol dm^{-3} hydrochloric acid with 10 g of marble chips (an excess of marble). In a second experiment the same procedure was used but with 100 cm^3 of 1.0 mol dm^{-3} acid and 10 g of the same size marble chips. The results of both experiments were plotted on a graph as shown in the diagram. Which line on the graph shows the results for the second experiment?

A A B B C C D D

Correct Answer: A B C D

Explanation:

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A student was investigating the reaction between marble chips and dilute hydrochloric acid.

Which changes would reduce the rate of reaction?

	temperature of acid	concentration of acid	surface area of marble chips
A	decrease	decrease	decrease
B	decrease	decrease	increase
C	increase	decrease	decrease
D	increase	increase	increase

Correct Answer: A B C D

Explanation:

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A student investigates the rate of reaction between magnesium and excess sulfuric acid. The volume of hydrogen given off in the reaction is measured over time. The graph shows the results of two experiments, R and S.

Which change in conditions would cause the difference between R and S?

A A catalyst is added in S.
 B The acid is more concentrated in R than in S.
 C The magnesium is less finely powdered in R than in S.
 D The temperature in R is lower than in S.

Correct Answer: A B C D

Explanation:

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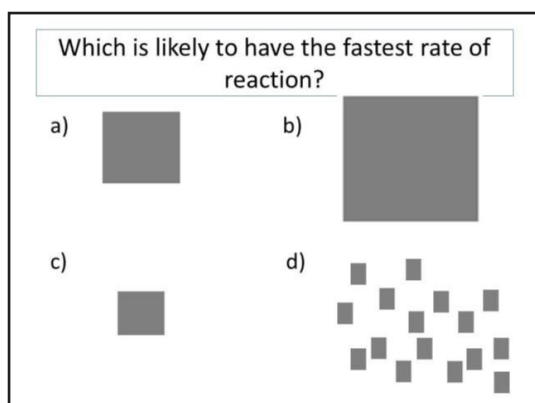
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Correct Answer: A B C D

Explanation:

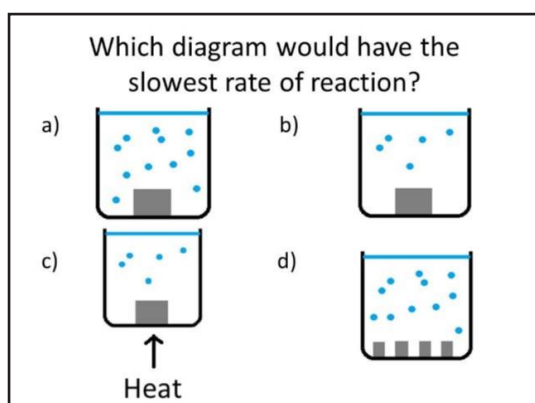
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Correct Answer: A B C D

Explanation:

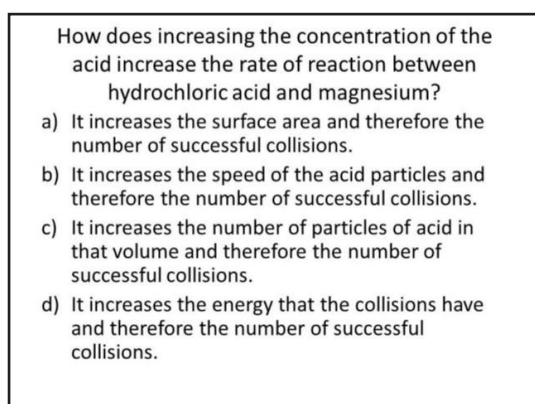
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Correct Answer: A B C D

Explanation:

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